

Subway Turnstile Controller Report

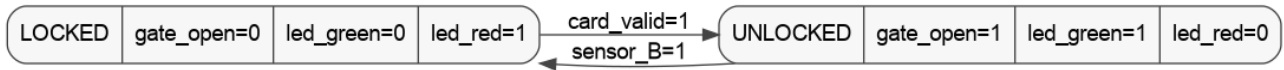
This report documents the Moore FSM design and simulation results for all four parts of the subway turnstile controller project. All Verilog modules were simulated with iverilog, and the waveforms below were generated from the resulting VCD files.

Design notes: asynchronous active-low reset is used in every module, sensor_B has priority over card_valid in the Part 3/4 IDLE state, and all delay logic uses the provided ALARM_DURATION and TIMEOUT_DURATION parameters.

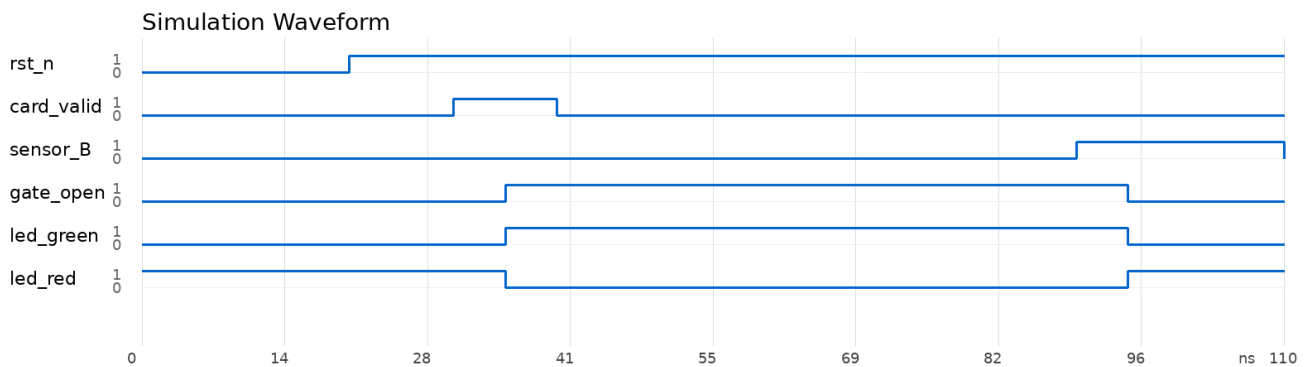
Part 1: Two-State Turnstile

A two-state Moore FSM is used. LOCKED keeps the gate closed with the red LED on. A valid card moves the controller to UNLOCKED, which opens the gate and enables passage. sensor_B closes the gate after the passenger exits.

State Diagram



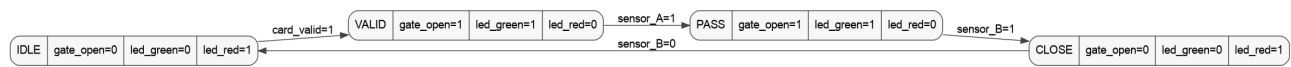
Simulation Waveform



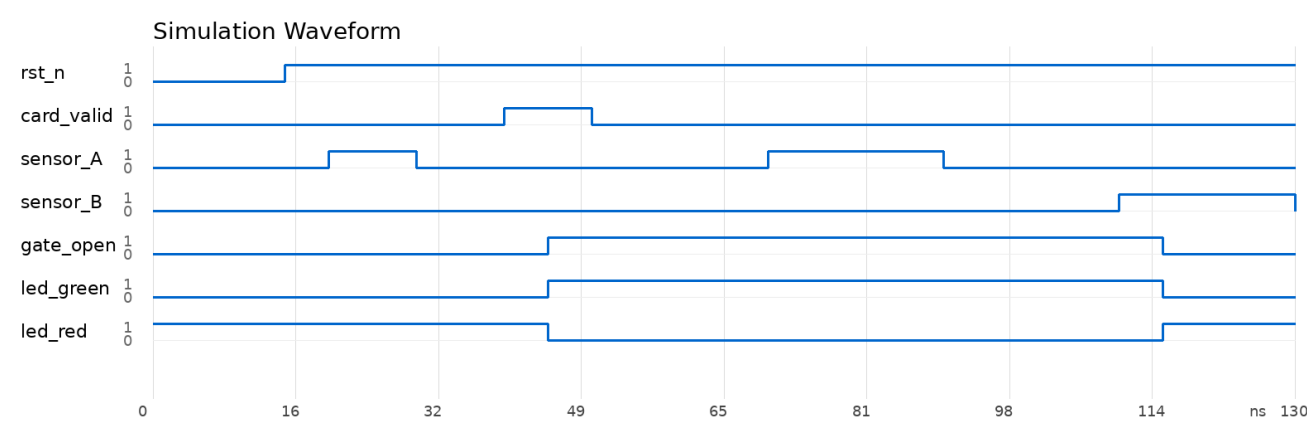
Part 2: Basic Access Control

The controller is extended to a four-state Moore FSM. IDLE waits for a card, VALID waits for lane entry at sensor_A, PASS waits for exit at sensor_B, and CLOSE holds the gate shut until sensor_B is released.

State Diagram



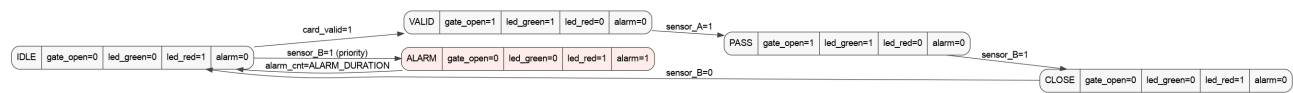
Simulation Waveform



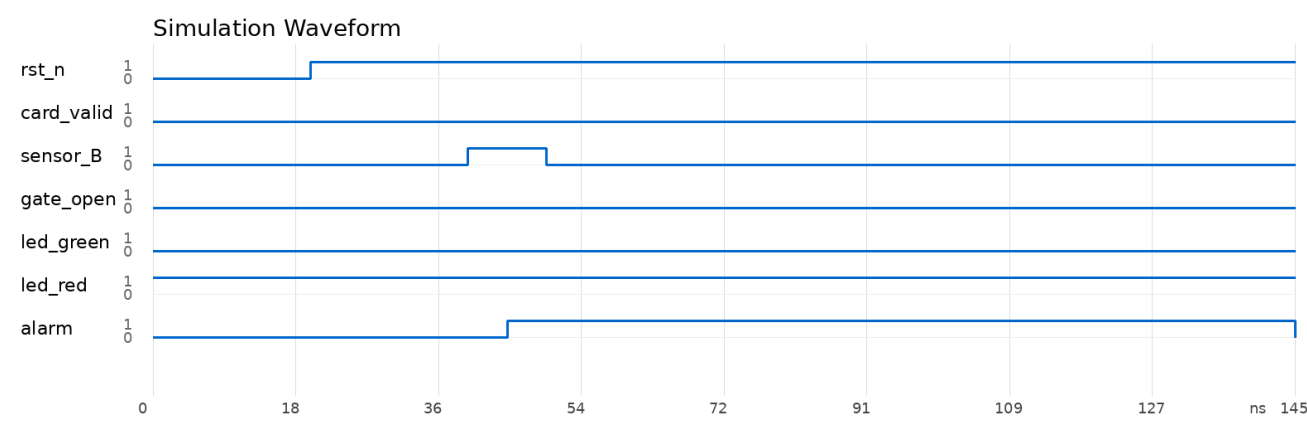
Part 3: Reverse Intrusion Detection

Part 3 adds an ALARM state and an alarm counter. In IDLE, sensor_B has higher priority than card_valid, so reverse intrusion is detected before any card action. The alarm stays high for ALARM_DURATION clock cycles, while the gate remains closed and the red LED remains on.

State Diagram



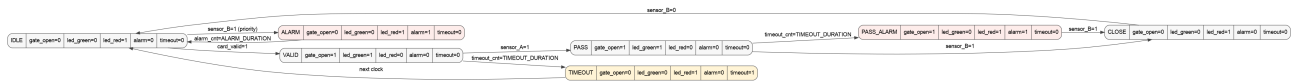
Simulation Waveform



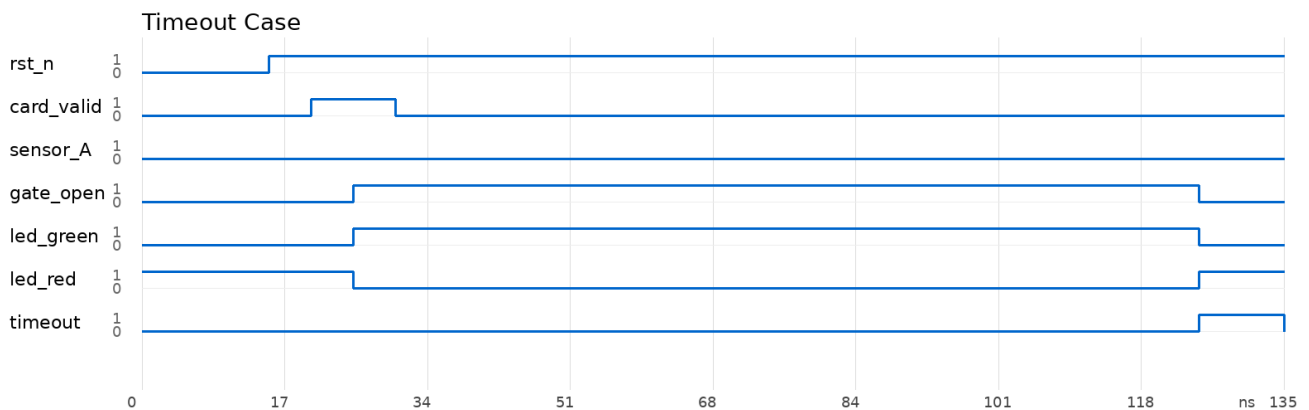
Part 4: Timeout Auto-Close

Part 4 keeps the reverse-intrusion alarm and introduces two more behaviors. In VALID, a timeout counter auto-closes the gate and asserts timeout when no passenger enters before TIMEOUT_DURATION. In PASS, the same timeout concept raises PASS_ALARM if a passenger stays in the lane too long. PASS_ALARM keeps the gate open, drives alarm high, and waits for sensor_B before finishing through CLOSE.

State Diagram



Timeout Case



Normal Passage Case

